S&DS 365 / 665

Intermediate Machine Learning

Reinforcement Learning: Policy Gradient Methods

November 4

Yale

Version of 2026/11/01 at 15:34

Automatic differentiation

- For supervised problems, loss function is $L(\widehat{Y}, Y)$
- Can program this directly
- RL loss functions are built up dynamically as agent makes decisions
- Automatic differentiation allows us to handle this

`https://www.tensorflow.org/guide/autodiff`

`https://pytorch.org/tutorials/beginner/basics/autogradqs_tutorial.html`

Automatic differentiation: Gradient collection

Modern platforms support automatic differentiation by recording relevant operations executed inside the context of a “tape”

This record is then used to compute the numerical values of gradients using “reverse mode differentiation”

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https://en.wikipedia.org/wiki/Automatic_differentiation

Automatic differentiation: Hello world!

```
import numpy as np
import torch
```

0.5s Python

```
x = torch.tensor(3.0, requires_grad=True)

y = x**2

y.backward()
dy_dx = x.grad
print(dy_dx.numpy())
```

0.0s Python

```
6.0
```

https://pytorch.org/tutorials/beginner/basics/autogradqs_tutorial.html

Automatic differentiation: Hello world!

```
w = torch.randn(3, 2, requires_grad=True)
b = torch.zeros(2, dtype=torch.float32, requires_grad=True)
x = torch.tensor([[1., 2., 3.]])

y = x @ w + b
loss = torch.mean(y**2)

[dloss_dw, dloss_db] = torch.autograd.grad(loss, [w, b])
print(dloss_dw.numpy(), "\n\n", dloss_db.numpy())
```

[3] ✓ 0.0s Python

```
[[ 0.63157207 -0.3194919 ]
 [ 1.2631441  -0.6389838 ]
 [ 1.8947163  -0.9584757 ]]

 [ 0.63157207 -0.3194919 ]
```

https://pytorch.org/tutorials/beginner/basics/autogradqs_tutorial.html

Multi-armed bandits



Multi-armed bandits

- The rewards are independent and noisy
- Arm k has expected payoff μ_k with variance σ_k^2 on each pull
- Each time step, pull an arm and observe the resulting reward
- Played often enough, can estimate mean reward of each arm
- What is the best policy?
- Exploration-exploitation tradeoff
- “Contextual bandits” add covariate vector — but actions don’t impact the environment

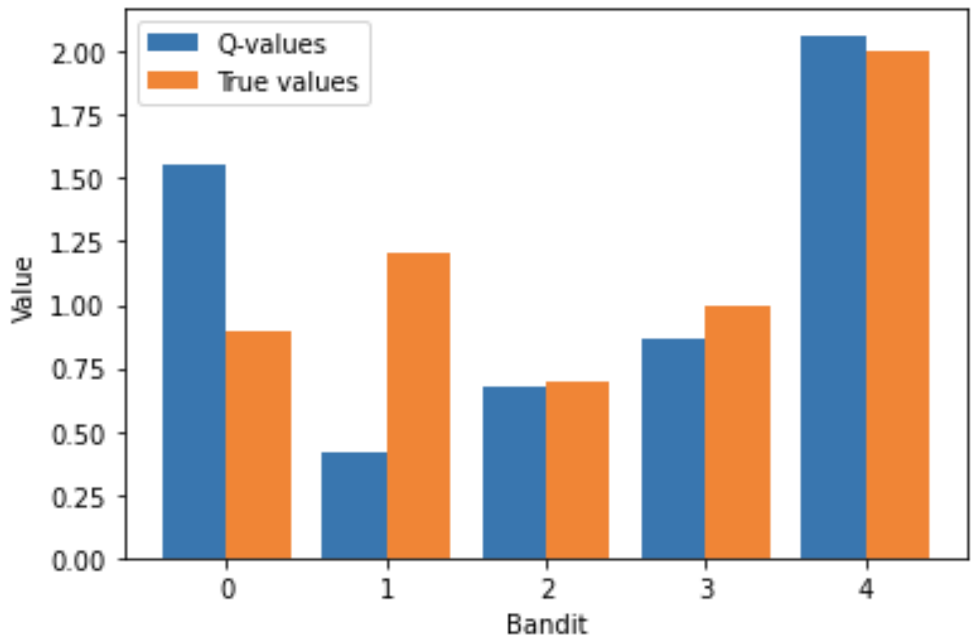
Multi-armed bandits

We'll treat this as an RL problem and hit it with a big hammer:
Deep Q-learning

Multi-armed bandits

```
======episode 10000 ======
Q-values ['1.556', '0.412', '0.675', '0.866', '2.065']
Deviation ['72.8%', '-65.7%', '-3.6%', '-13.4%', '3.3%']

<Figure size 864x504 with 0 Axes>
```

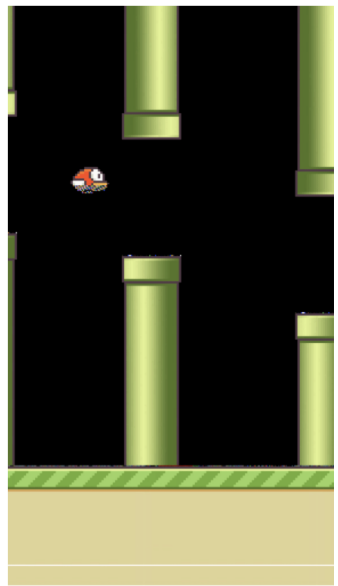


Let’s go to the notebook!

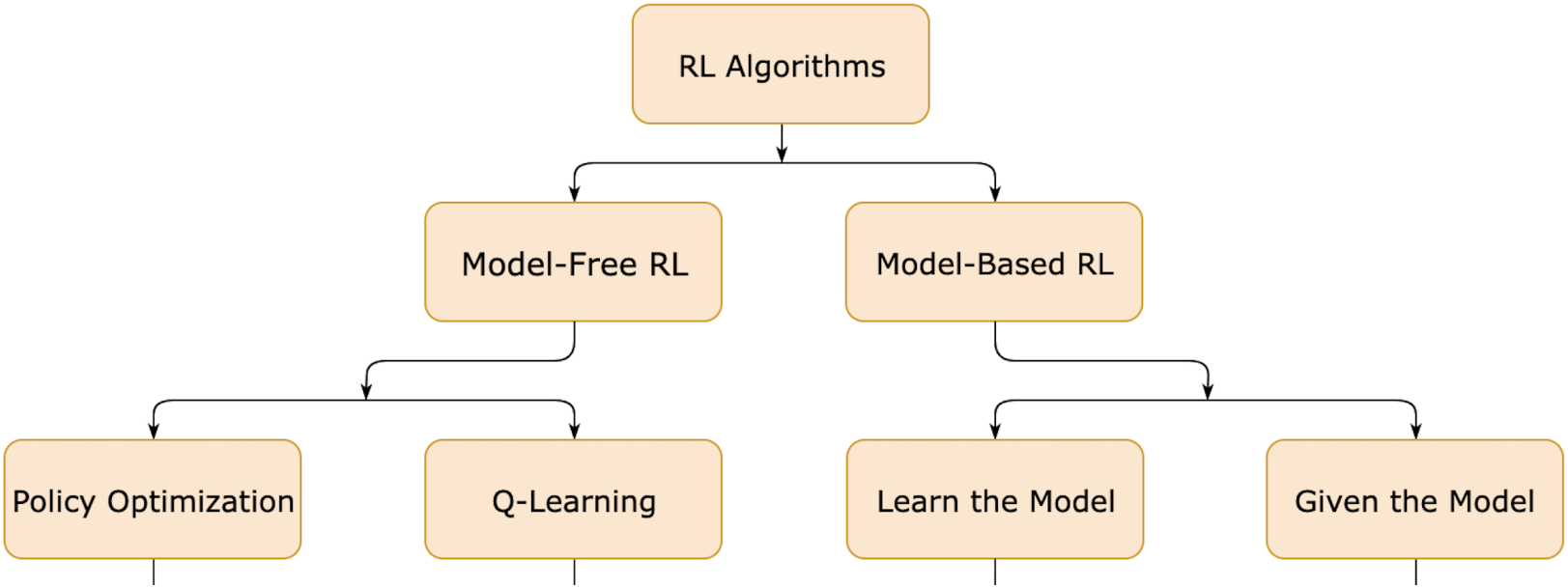
Assn 4: Flappy Bird

Problem 3: Deep Q-Learning for Flappy Bird (25 points)

In this problem, we will walk you through the implementation of deep Q learning to learn to play the Flappy Bird game.



Landscape of RL algorithms

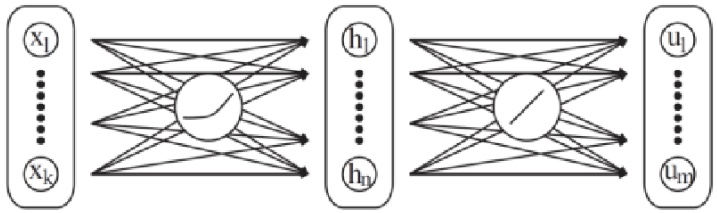


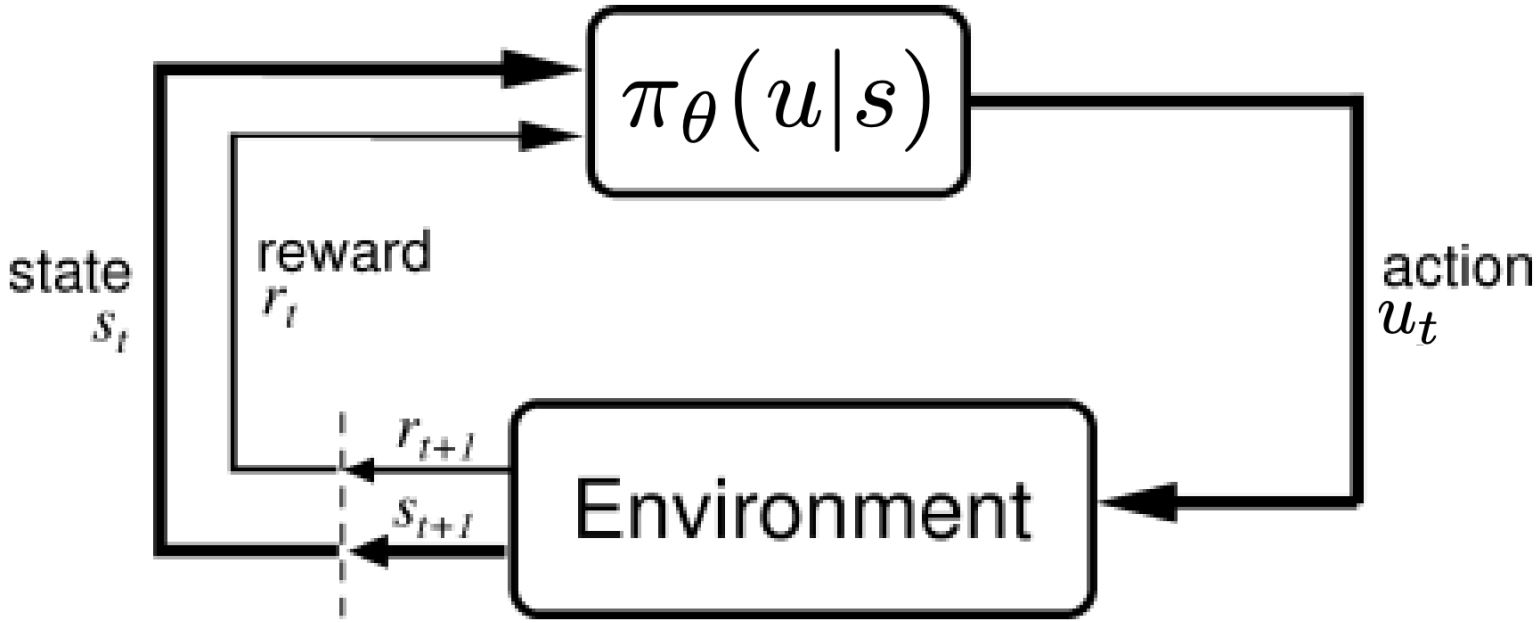
https://tinyurl.com/25vruwtp

Policy gradient methods

- policy $\pi_\theta(s)$ has parameters θ
- Perform gradient ascent over θ
- Well-suited to deep learning approaches
- Why use an on-policy method? May be possible to estimate a good policy without accurately estimating the value function or Q-function
- Can be more compute-efficient and stable than Q-learning

Policy gradient methods





Successes of policy methods



Kohl and Stone, 2004



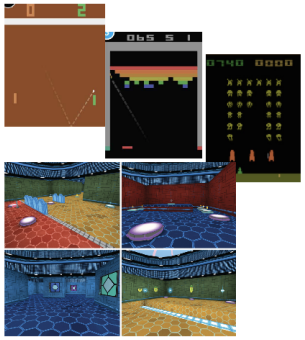
Ng et al, 2004



Tedrake et al, 2005



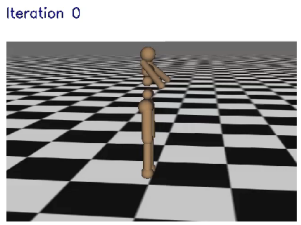
Kober and Peters, 2009



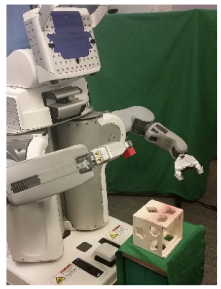
Mnih et al, 2015
(A3C)



Silver et al, 2014
(DPG)
Lillicrap et al, 2015
(DDPG)



Schulman et al,
2016 (TRPO + GAE)



Levine*, Finn*, et
al, 2016
(GPS)



Silver*, Huang*, et
al, 2016
(AlphaGo**)

Policy gradient methods: Loss function

We start with the loss function: Expected reward $\mathcal{J}(\theta) = \mathbb{E}(R)$

- Policy $\pi(s; \theta)$ has parameters θ
- Perform gradient ascent on $\mathcal{J}(\theta)$
- Well-suited to deep learning approaches
- Neural network policy extracts predictive "features" of states

We'll need to develop some mathematics to understand how policy methods can be efficiently applied...

Policy gradient methods: Loss function

Policy is probability distribution $\pi_\theta(a \mid s)$ over actions given state s.

The episode unfolds as a random sequence τ

$$\tau : (s_0, a_0) \to (s_1, r_1, a_1) \to (s_2, r_2, a_2) \to \cdots \to (s_T, r_T, a_T) \to s_{T+1}$$

where s_{T+1} is a terminal state. Receive reward $R(\tau)$, for example

$$R(\tau) = \sum_{t=1}^{T} r_t$$

Objective function $\mathcal{J}$ is expected reward

$$\mathcal{J}(\theta) = \mathbb{E}_\theta(R(\tau))$$

Calculating the gradient

Using Markov property, calculate $\mathbb{E}_\theta(R(\tau))$ as

$$\mathbb{E}_\theta(R(\tau)) = \int p(\tau \mid \theta) R(\tau)\, d\tau$$

$$p(\tau \mid \theta) = \prod_{t=0}^{T} \pi_\theta(a_t \mid s_t)\, p(s_{t+1}, r_{t+1} \mid s_t, a_t)$$

If states or rewards are finite the integral becomes a sum, or a mix of sums and integrals.

Calculating the gradient

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It follows that

$$\nabla_\theta \log p(\tau \mid \theta) = \sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t \mid s_t) = \sum_{t=0}^{T} \frac{\nabla_\theta \pi_\theta(a_t \mid s_t)}{\pi_\theta(a_t \mid s_t)}$$

If states or rewards are finite the integral becomes a sum, or a mix of sums and integrals.

Calculating the gradient

Now we use

$$
\begin{aligned}
\nabla_\theta \mathcal{J}(\theta) &= \nabla_\theta \, \mathbb{E}_\theta R(\tau) \\
&= \nabla_\theta \int R(\tau) \; p(\tau \,|\, \theta) \; d\tau \\
&= \int R(\tau) \, \nabla_\theta p(\tau \,|\, \theta) \; d\tau \\
&= \int R(\tau) \, \frac{\nabla_\theta p(\tau \,|\, \theta)}{p(\tau \,|\, \theta)} \; p(\tau \,|\, \theta) \; d\tau \\
&= \mathbb{E}_\theta \Big(R(\tau) \nabla_\theta \log p(\tau \,|\, \theta) \Big)
\end{aligned}
$$

Need regularity conditions to exchange derivative and integral, such as uniform continuity of the derivatives of the policy.

Approximating the gradient

Since it's now an expectation, can approximate by sampling:

$$\begin{aligned}
\nabla_\theta \mathcal{J}(\theta) &\approx \frac{1}{N} \sum_{i=1}^{N} R(\tau^{(i)}) \nabla_\theta \log p(\tau^{(i)} \mid \theta) \\
&= \frac{1}{N} \sum_{i=1}^{N} R(\tau^{(i)}) \sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t^{(i)} \mid s_t^{(i)}) \\
&\equiv \widehat{\nabla_\theta \mathcal{J}(\theta)}
\end{aligned}$$

Approximating the gradient

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$$\begin{aligned}\nabla_\theta \mathcal{J}(\theta) &\approx \frac{1}{N}\sum_{i=1}^{N} R(\tau^{(i)})\nabla_\theta \log p(\tau^{(i)} \mid \theta) \\ &= \frac{1}{N}\sum_{i=1}^{N} R(\tau^{(i)}) \sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(a_t^{(i)} \mid s_t^{(i)}) \\ &\equiv \widehat{\nabla_\theta \mathcal{J}(\theta)}\end{aligned}$$

$\tau^{(i)}$ is sometimes called a "rollout" of the policy π_θ

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The policy gradient algorithm is then

$$\theta \longleftarrow \theta + \eta \widehat{\nabla_\theta \mathcal{J}(\theta)}$$

Intuition

Gradient ascent of the expected reward with respect to policy:

- Increases probability of paths τ with large R
- Decreases probability of paths τ with small R

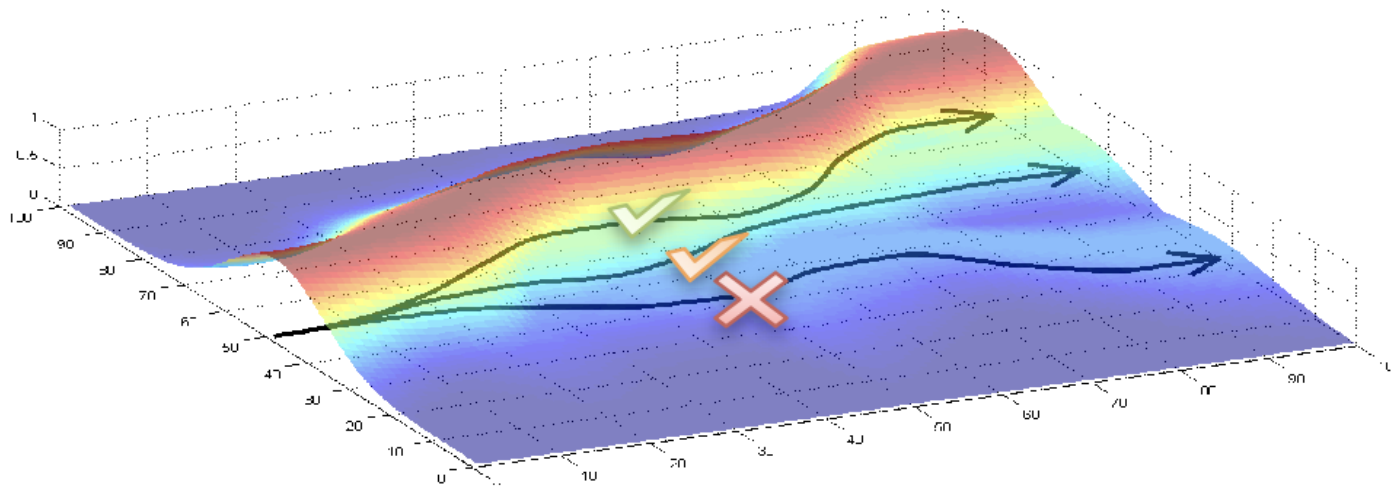
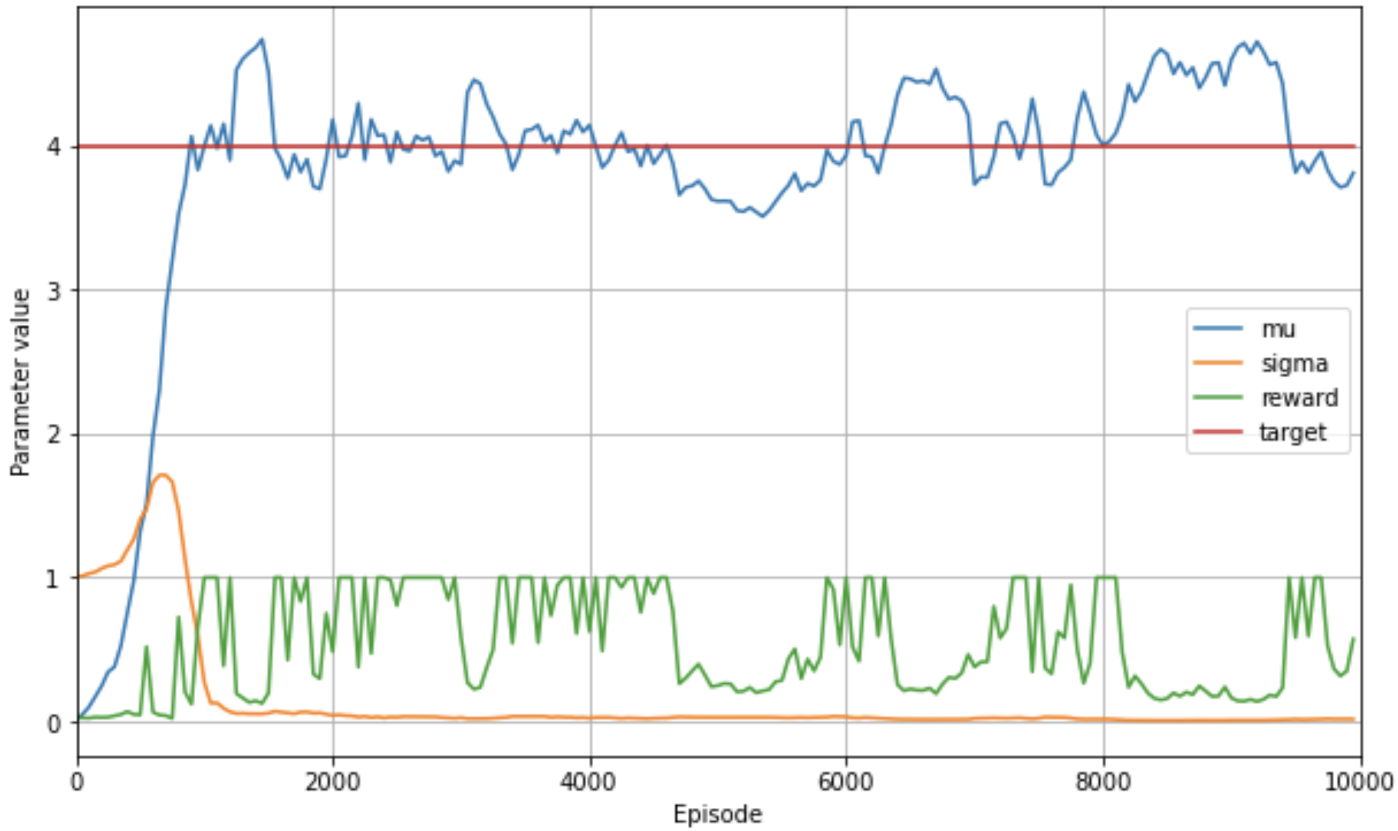


Fig from Schulman and Abbeel

Simple example



Simple example



Let’s go to the notebook

Summary so far

- Policy methods estimate $\pi(a \mid s)$
- Tabular methods can be used for small state/action spaces
- Otherwise, parameterize $\pi_\theta(a \mid s)$ and use gradient ascent
  - Change in expected reward calculated with “grad-log trick”